

 Main Guide



A Guide To Hackathons

by Simonee Ezekiel Mariquit AKA Stimmie

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Join the hackathon pool [here](#)

This document is mostly tailored to an audience of young talent, students and new grads. in the Philippines.

If you're mostly looking for [resources](#), [samples](#), or [my experience](#), see the appropriate section of this document.

This is a living document. VERSION: 0.0.1



Hi! I'm Stimmie and I like to join hackathons, among other interests. As of 2025, I've joined **24** and deeply enjoyed each one of them, at least in hindsight. I've teamed with **66 people** and got to know **hundreds of people** in this space.

I've researched about the problems of **10+ fields** and really learned a lot. My undying passion in tech, entrepreneurship, and especially mentorship makes hackathons a perfect space for me to grow.

[Major League Hacking](#) defines a hackathon as follows:

“an event where participants learn, build, and share their creations over a weekend in a relaxed, welcoming atmosphere”

However, I like [Joshua Tauberer's](#) definition a bit more:

“Hacking is creative problem solving. (It does not have to involve technology.) A hackathon is any event of any duration where people come together to solve problems. Most hackathons I've run also have a parallel track for workshops”

What are hackathons?

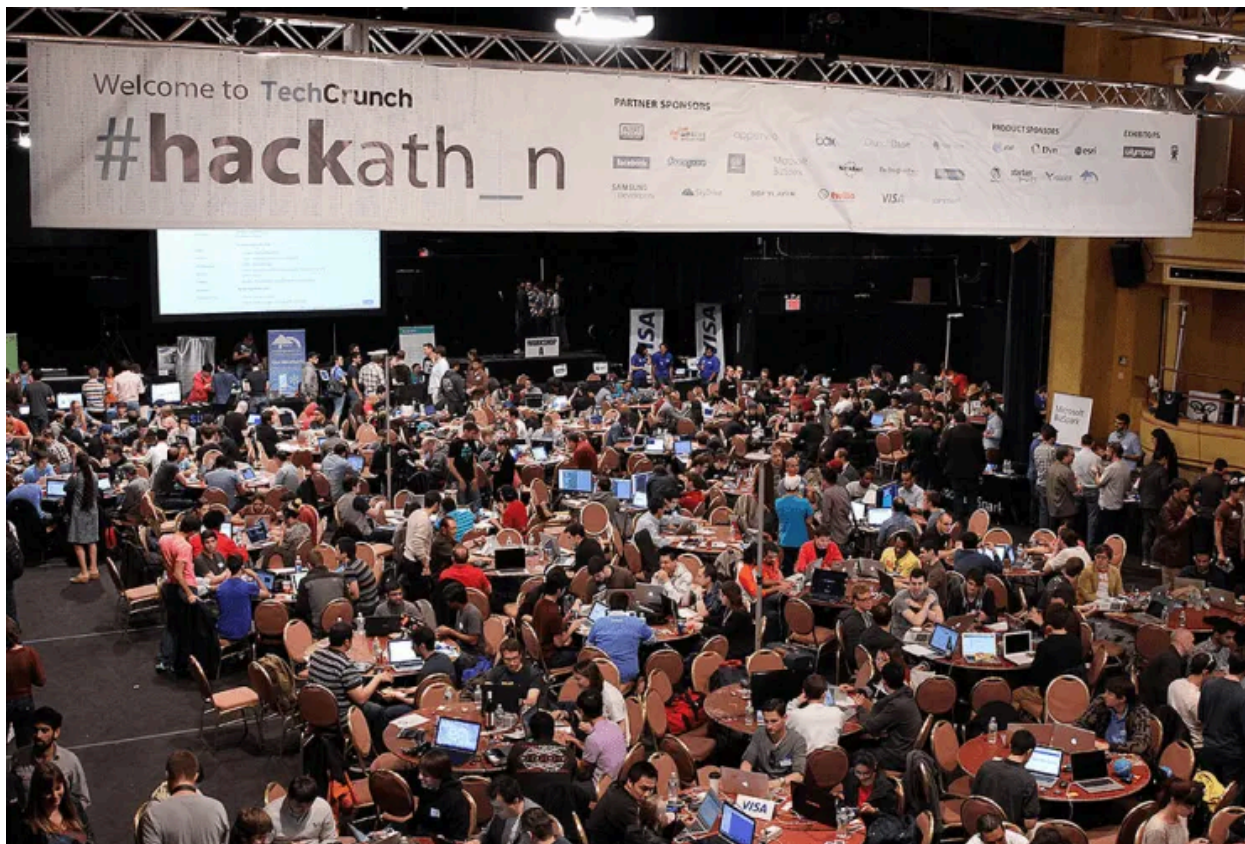


Image by Active Mission

Basically, it's an invention marathon. "Permissionless innovation". The software part of it is deliberately not part of the definition because **hackathons are not merely for CS/IT/CpE/<insert tech field>/etc. students**, although most of its participants do indeed come from that demographic.

A hackathon can be of any duration and modality. Some hackathons happen face-to-face at a venue like a function hall or a company building. Some are hybrid, having you come to the venue for the orientation and the pitching/awarding. A lot of them are fully virtual, allowing you to create stuff from anywhere and get to know a diverse set of people.

For instance, here are examples of how some programs help with the general spirit of product-focused problem solving:

- Psychology helps you understand how users feel about the product and why they behave the way they do (user motivation, cognitive load, decision-making biases) to design intuitive user experiences (UX).
- Engineering focuses on feasibility, physical implementation, and efficiency. People in engineering are inventors by profession.
- Business drives the viability and market strategy of the product. They know if the product solves a real user need.
- Arts brings expertise in aesthetics, communication, and emotional connection
- Medical fields focus on safety, ethical impact, and genuine human welfare.

To an extent, I agree that solutions made in hackathons tend to never make a real-world impact. The value of hackathons, to me, comes with the transferable skills: problem-solving, critical thinking, project management, playing with a team, networking, selling a product, and other hard skills.

Real world problems are hard and don't get solved in a weekend by a bunch of sleep-deprived young adults!

For the purposes of this document, we'll define a "hackathon" as including not only software sprints, but also:

- ✓ Ideathons, where you make a product idea and pitch it.
- ✓ Hackathons that involve hardware aspects, especially Internet-of-Things (IoT)
- ✓ "Buildathons", just another word for hackathon

- ✓ Design competitions, focusing on using Figma, Adobe XD, or other prototyping software
- ✓ Game jams, focusing on full game development, including its art aspects
- ✓ Data challenges, usually involving finding insights on a dataset or theme.

Meanwhile, these will not fall under “hackathons” for this document

- ✗ Competitive programming, focusing on math and algorithms (DSA)
- ✗ Quiz competitions
- ✗ Capture the flag (CTFs), focusing on finding security breaches.

Why join hackathons?

Not all hackathons are made equally. Some are poorly organized, if not intentionally dubious. However, when organized properly, hackathons can be a meaningful **exercise** in problem-solving. I emphasize the word “exercise” because most of the time, you won’t be solving a real problem on the implementation-level and that’s okay.

The value often lies not in producing a finished product, but in the *process*: ideating, collaborating, experimenting, and learning under pressure.

In contrast to professional software development, where timelines are long and stakes are high, hackathons offer a low-risk space to try bold ideas. There's freedom in knowing that failure is expected and even welcomed and encouraged :).

I can’t code/sell/pitch/etc. + Wala akong experience

▶ [Hacking Hackathons: how everyone can contribute to innovation | Angela ...](#)

The video above is a TED talk telling the story of joining a first hackathon to join up to 50. Personally, it’s a nice watch.

▶ [Confessions of a Hackathon Addict | Chris Chan | TEDxSFState](#)

▶ [The power of hackathons | Krithika Yetchina | TEDxRedmond](#)

▶ [Hack your life in 48 hours | Dave Fontenot | TEDxTeen](#)

Above are some other TED talks about hackathons.

Inexperience is one of the most common reasons why people hesitate to join hackathons, and understandably so. A lot of people hold back because of [imposter syndrome](#). **I'll let you in on a secret: you can win some hackathons with only Canva slides and a nice pitch.**

In my first hackathon, I didn't know how to code a web app. My expertise was mostly data structures and algorithms and I had only fiddled with mostly Java, which basically no one uses for hackathons.

Great teams are built on diverse strengths. Most of the time, one developer/technical person will suffice because, again, hackathons are an exercise in problem-solving. So long as you know how to solve problems, you can join a hackathon.

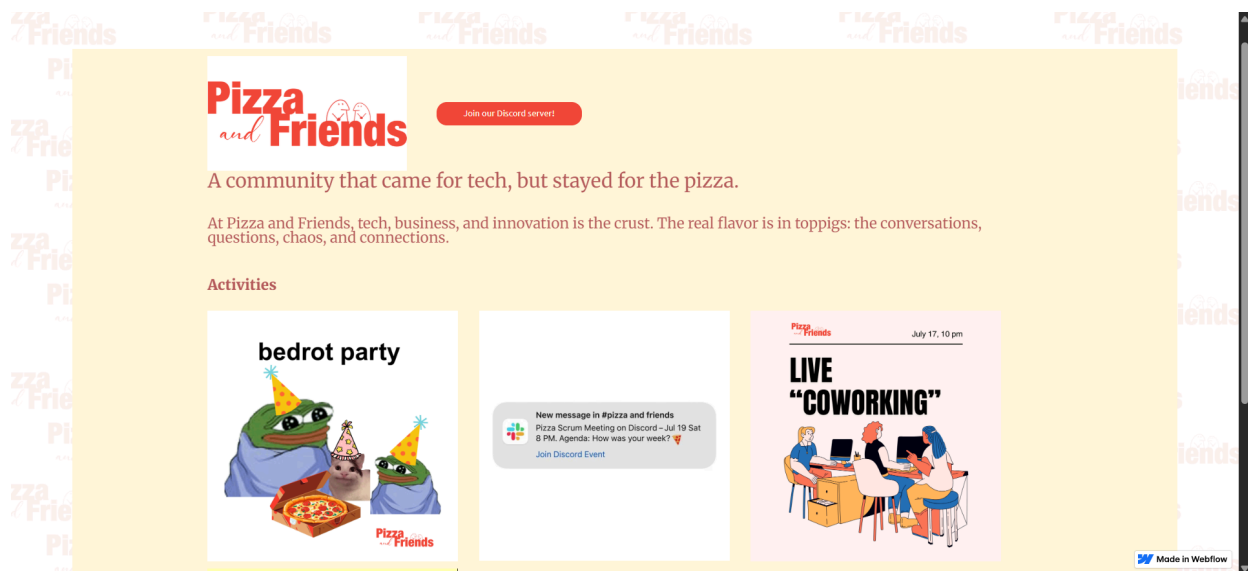
Hackathons aren't about showcasing perfection; they're about showing up and being willing to figure things out together. If you walk in with curiosity and a willingness to contribute, you'll walk out having learned something new—about the process, your teammates, and yourself.

Who should I join with?

Anyone! I've personally benefitted from joining with strangers, learning many different perspectives along the way. Most of them I've enjoyed working with, some of them not so much, but I've learned really not to take it too personally. You may join with friends, classmates, orgmates, workmates as you already have established dynamics. You can also ask the organizer to assign you to a team! Just consider how much the team jives together as this would greatly affect your success.

Okay, I'll join one!

Great! Now, you'll first have to find a hackathon. You can start with checking the Facebook pages of orgs in your university/college, or the one you graduated from. You'll probably have luck if the org is about science and technology or entrepreneurship.



Pizza and Friends website

I also happen to [run a community](#) that's quite related to tech, and we sometimes send opportunities there.

Keep in mind that a lot of hackathons only allow enrolled students to join. If you're not enrolled, you can probably skip the step below:

If there are none, you can start following tech-adjacent organizations in other universities, especially bigger ones, such as UP, Ateneo, La Salle, PUP, PLM, FEU, just to name a few. Some universities also have Google, AWS, and [insert tech company] clubs, which also sometimes host hackathons.

A lot of hackathons are also organized by non-student organizations. A simple search of "hackathon" and related keywords, most of the time, suffices. This [Facebook Group](#) posts some opportunities.

Lastly, if you want to try your hand in international hackathons, try checking [Devpost](#) and [Major League Hacking or MLH](#). If you want to join Game Jams, check [Itch.io](#). For blockchain-focused ones, check [DoraHacks](#)

Hackathon Skills

You'll need a team that has a diverse skillset. You might need to wear multiple hats, but here are some roles that you can fill.

- Developers (tech) - make the actual product or prototype

- Designers - make UI/UX design, mockups, and design systems
- Researchers - validate the problem and solution, conduct user research
- Content - create the copy (writing), documentation, pitch scripts, pitch deck, etc.
- Pitchers - deliver the final pitch to the judges
- Project Management - make sure the project gets done

Bring whatever skills you currently have and learn the gaps along the way.

About “Good Code”

When you use apps on your phone or play an enjoyable game, do you ever stop to appreciate how *well-coded* this piece of software is? Most people would say no.

As users, we mostly just think about the experience, as in how this product makes us feel. With this in mind, while good code is useful for production systems and other use cases, if you simply want to demonstrate a software idea, there is no need for “good code”.

Recommended tools

- **Version Control System:** GitHub
- **Communication Tool:** Discord, Telegram, Messenger, whatever
- **Project Management Tools:** Trello, Asana, Notion, Google Sheets/Docs
- **Cloud Storage:** Google Drive
- **Slides:** Canva or Figma

Understand the problem first!

For this section, credits go to the workshop by [Richard Day](#), which was organized by [START - DOST](#).

All solutions start with your users. The most common mistake is going straight to making your product before doing user research.

User Research Questions:

- “Can you describe your typical transaction with X?”
- “What frustrates you most about doing X?”

- “How would you describe a *perfect* version of this experience?”

Avoid "ice cream questions" (e.g., “Would you like this feature?”) that lead people to say yes without thinking. Focus on past behavior and real frustrations. Ask for **stories**, not opinions.

You may also conduct an interview. Make sure to try having the following roles filled:

- **Lead interviewer:** asks questions
- **Secondary interviewer:** follows up or supports
- **Note-taker:** captures insights
- **Catch-all:** observes tone, body language, etc.

Use **How Might We** or **In What Ways Might We** questions to reframe problems into opportunities.

Example:

- Problem: *Drivers are canceling rides in bad weather.*
- How might we make it easier for drivers to accept rides during rain?

Look for Problems That Are:

1. Popular – lots of people have this issue
2. Growing – it's becoming more relevant
3. Urgent – needs solving now
4. Expensive – costs time, money, or energy
5. Mandatory – no one can avoid it
6. Frequent – happens often, e.g. multiple times a day

Think of a solution

Now that you understand the problem properly, you can **avoid a solution looking for a problem**. It's time to brainstorm ideas. Here's a recommended framework to do so:

- Create a focus question before starting.
- Set a time limit (15–30 minutes).
- Assign a facilitator to guide the session.
- Encourage a judgment-free, open environment.

Brainstorming Techniques:

- Wrong Answers Only – encourages creative, unfiltered ideas.
- SCAMPER – Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse.
- MoSCoW – Must-have, Should-have, Could-have, Won't-have
- Visual tools – use sticky notes, whiteboards, or online canvases like Miro or FigJam.
- [Artificial Constraints](#)
- [10x10](#)
- [6-3-5](#)
- [Rolestorming](#)
- Create [user personas](#)
- Keep in mind the [double diamond framework](#), although it has its [critiques](#).
- Downselect the ideas

You may use the SE (Service Experience) Framework to outline how a user interacts with your product or solution:

1. Entice – how users discover your solution
2. Enter – the first interaction
3. Engage – the main experience
4. Exit – how they complete or leave the session
5. Extend – how you bring them back (retention, referrals)

AIDA Framework (Marketing & Pitches)

Use this to structure your user pitch, landing page, or product presentation during or after the hackathon.

1. Attention – Grab their focus (a stat, a question, a pain point).
2. Interest – Explain the value in a way that resonates.
3. Desire – Make it feel personally relevant and emotionally compelling.
4. Action – Ask for a clear next step (download, sign up, vote, invest).

5 Whys Technique (Root Cause Analysis)

Use the 5 Whys to dig deeper into a user problem or frustration. Start with a problem and keep asking “Why?” until you reach the root cause. You can ask 4, 6, or 7 times as you please, so long as you find the root cause of the issue.

Example:

- *Users are abandoning the checkout page.*
 - Why? Because it takes too long to load.
 - Why? Because it has too many fields.
 - Why? Because we ask for info not needed immediately.
 - Why? Because we copied a desktop form into mobile.
 - Why? Because we didn't design with mobile-first in mind.

Making the product

When you've identified what you want to do, you should decide the specific tools you will use for this. As much as you should use the right tools for the job, you should probably use a tool you're already good at. The set of tools that you will use is called the [Tech Stack](#).

Your solution does not need to be complete. It does not necessarily need to even be *functional* (though bonus points if you do!). Your priority should be to make something that *communicates* the ideal version of what you want to do, and it should at least have the core user flow. This is called a [Minimum Viable Product](#).

Here are some features you should not waste time on:

- Authentication/Login
- Robust database setup
- Advanced error logging
- Email notifications
- Complex user settings

Presenting the product



One of my co-founders of Pizza and Friends, John, pitches our app at the Byte Forward Hackathon.

When presenting the product,

Now, you'll also have to demo the actual product. Ideally, you'll actually run the product on your computer.

- Only demo the features that are functional and essential to the core problem you are solving.
- Use pre-loaded data to prevent slow loading or unpredictable network errors,
- Always have a backup plan, such as a recorded video of the demo or high-fidelity screenshots in your deck, in case of technical failure.

Other things you could do during and after

You shouldn't just be there at the hackathon to actually work on your thing and win. In my opinion, that's being too serious. You're in an environment with many people with the same interests, this is the *perfect* opportunity to connect with people! **Don't be afraid to talk to other teams!** Look at other people's solutions, give feedback, and ask for feedback in your work. This is also an opportunity to explore the vicinity of the venue, buy some souvenirs, and go to the local spots.

After the hackathon, don't forget to celebrate whether you won or not! The glory is always in the attempt. This is also an opportunity to publish your project on platforms like GitHub or a social media app of your choice. You may also synthesize everything you learned and debrief with your team about what went well and what can be improved.

My critique of hackathons

As I mentioned in the beginning, not all hackathons are built equally. Hackathons aren't perfect, but as with everything that attempts to emulate the real world, there is no perfect simulation. Even then, actually making something is better than just watching an endless stream of YouTube tutorials.

1. Winners and Losers

I don't really like the entire system of hackathons having winners and losers. It fosters a competitive environment and makes the success of people contingent on the minds of three or more people, when in reality millions of people that comprise a market determine a product's success.

2. Crunch Culture

Some hackathons load you with caffeine and junk food and make you stay up all night, which I believe lessens the effectiveness of a project.

3. Unpaid Labor

Too often, companies use hackathons as a glorified way to get free labor and ideas from unsuspecting students.

Synthesis

Overall, hackathons are a cool way to learn stuff, make ideas, and meet new people and mentors. It's a good enough simulation of problem solving, despite not fully emulating the real world at times. Join one at least once in your life :)



My Hackathon History



At A Glance

- With the definition in the [main guide](#), I've joined **24** hackathons.
 - 3 of them are 24-hour hackathons
 - More than half of them have are hybrid or face-to-face
- I have made **thirteen** software outputs (as opposed to prototypes or just slides)
- I have **championed** three of those hackathons.
- Including the three championships, I have podium'ed six times.
- I have worked with **66** different people.
- I have personally brought home **28,383 Php** worth of prize money
- As a team, we've won **131,500 Php**.
- I've competed with 500+ other teams, which is at least **2,000** people.
- I've connected with **more than 100** mentors and organizers
- I've gone to more than **20 new places** from joining hackathons.
- I've explored more than **15 fields** in my projects
 - a. Education
 - b. Agriculture
 - c. Civic Tech
 - d. Public Health
 - e. Blockchain
 - f. Fintech
 - g. Policy
 - h. Sustainability
 - i. ESG
 - j. Safer Cities
 - k. Atmospheric Resilience
 - l. Game Development
 - m. Digitalization
 - n. Literacy

- o. Electric Vehicles
- p. etc.
- I've listened to **100+ pitches**, seen more than **500 ideas**, most of them shitty, some of them brilliant.

Every Hackathon I've Joined (WIP)

(Chronological Order)

1. GDSC PLM Innolympics (24h Hackathon)
 - a. <https://devpost.com/software/iskosenso>
2. UPLB ACSS Internal Ideathon - Champion
 - a. <https://projects.stimmie.dev/dinggin>
- UPLB CPAF Data 2 Decision: Data Visualization Contest - Champion (3,000 Php)
 - a. <https://projects.stimmie.dev/agri-infographics>
- GDG on Campus - UP Manila GenAlus Hackathon - 2nd Runner-up (500 Php)
 - a. <https://projects.stimmie.dev/civik-ai>
- START - DOST National Hackathon - 2nd Runner-up (10,000 Php)
 - a. <https://projects.stimmie.dev/smile-connect>
- Internet Computer Protocol (ICP) Blockchain - iThink Hackathon 2025 - 2nd Runner-up (10,000 Php)
 - a. <https://projects.stimmie.dev/skillmint>
- APRU Tech Policy Hackathon - Shortlisted
 - a. <https://projects.stimmie.dev/tiwala>
- UP Los Baños - YSES: Future Ideator Quest
 - a. <https://projects.stimmie.dev/adventurine>
- World Computer Hacker League (ongoing) - Regional Round Qualifier
 - a. <https://projects.stimmie.dev/skillmint>
- FEU-CPeO - Create and Conquer
 - a. <https://projects.stimmie.dev/crewmate>
- BPI DATA Wave 2025
 - a. <https://projects.stimmie.dev/vybe-change>
- UPLB COSS Web Design Competition
 - a. <https://projects.stimmie.dev/unawa>
- PUP Spark Fest
 - a. <https://project-illumina.netlify.app/>
- Los Banos Game Jam
 - a. <https://jlscsr.itch.io/fwdpeers-filipinolympics>

- Philippine Junior Data Science Challenge
- Byte Forward Hackathon

My Story (WIP)



Samples

Samples

This section provides a comprehensive overview of sample outputs from various hackathon projects, categorized for easy navigation. Explore the diverse range of work, from functional software products to compelling pitch materials and interactive prototypes.

Software Products

This subsection showcases functional software applications developed during hackathons. Each entry includes a link to the GitHub repository and, where available, a live website.

Pitch Decks

This subsection features various pitch decks used to present hackathon projects. Each entry includes a link to the design platform where the pitch deck is hosted.

Pitch Videos

This subsection presents sample pitch videos created for hackathon projects, offering a dynamic overview of the presented solutions. Each entry includes a link to the video file.

Prototypes

This subsection features interactive prototypes developed during hackathons, providing a visual representation of user interfaces and functionalities. Each entry includes a link to the prototype.



Software Products

Sample Software Outputs

SkillMint

 smmariquit/SkillMint | [Website](#)

Illumina

 smmariquit/illumina | [Website](#)

LagunaGrown

 smmariquit/laguna-grown

Filipinolympics

 API Error

CivikAI

 smmariquit/civikai | [Website](#)

[Not Mine]

<https://overdrivengain.github.io/cs132-portfolio/#portfolio>

<https://interactive.theguidon.com/>



Pitch Decks

Sample Pitch Decks

SkillMint

 Canva

TravelBuddy

 Canva

Adventurine

 Canva

Dinggin

 Canva

Takbilis AI

 Canva

CivikAI

[Figma Slides](#)

[Not Mine]

 (Reference) ASDigi Pitch Deck.pdf

 (Reference) FlexLife Pitch Deck.pdf



Pitch Videos

Sample Pitch Videos

SkillMint

 FINAL - SkillMint Video Pitch.mp4

AquaGuard

  AquaGuard Explainer - IoTayo UPLB.mp4

Illumina:

 Illumina - Demo Video.mp4

[Not Mine]

<https://www.youtube.com/watch?app=desktop&v=Y8IxKMnsaHw&t=2947s>

Prototypes

Sample Prototypes

SkillMint

 SkillMint

Game Jam

 Space Jammmm

Adventurine

 Untitled



Briefers



<https://mangrove-kettledrum-f3d.notion.site/Iskosenso-MangioneGPT-178423a0182c800db1cef522c4110a87>

<https://devpost.com/software/iskosenso>

<https://innolympics-2025-gdsc-plm.devpost.com/>

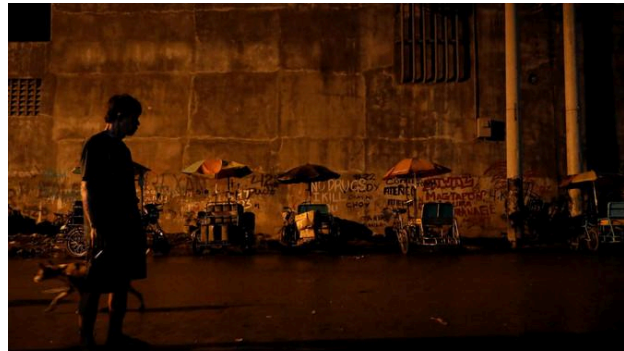


Illumina

Project **Illumina**: Designing Cities for People

project-illumina.netlify.app | [Illumina APK](#) | github.com/smmariquit/illumina | [Figma](#)

A computer vision (CV) powered solution to light up our cities. Empowering SDG [11](#) & [16](#).



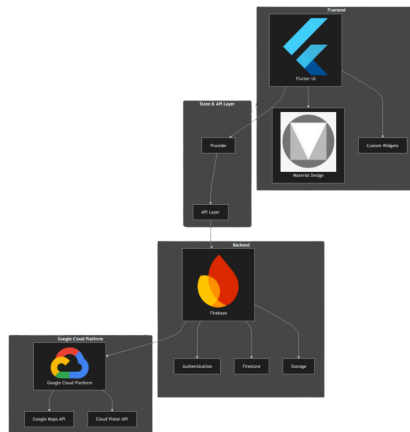
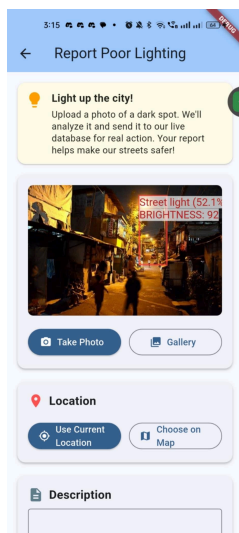
A suffocating gloom clings to the streets, swallowing even the echoes of desperation in Manila's forgotten corners.

Sometimes, urban crime is encouraged by fundamental design flaws. Poor street lighting, lack of visibility in public spaces, isolated walkways, dead-end alleys, and inadequate surveillance due to high walls or blind corners, to name a few. In the Philippines, particularly in Metro Manila, this contributes to not only a high crime rate, but also poses general hazards to Filipino citizens.

Illumina is Phase 1 of the **Design Cities for People Initiative (DCPI)**—a civic tech project that combines **real-time data**, **computer vision**, and **community** reporting to create safer urban environments. The mobile app maps danger zones, highlights poorly lit or high-crime areas, and provides live alerts. Users can report

hazards, share safety updates, and notify trusted contacts when entering risky zones. As the first step in DCPI, **Illumina** lays the groundwork for future tools that use computer vision and crowd-sensing to tackle deeper urban design challenges. **Powered by Google technologies** as a proof-of-concept, is a cross-platform **Flutter** app hooked into **Firestore** for storage, database, and authentication, and **Google Cloud Platform** for its mapping and vision functionality.

For now, **Illumina** helps detect 1) places where the local government unit (LGU) can focus its urban infrastructure projects, giving crucial data to city engineers, and 2) gives users insights on safety hazards, especially at night. Scoping out, once the



DCPI project gets its hands on funding, larger datasets, and better-trained models, this project can be more than an app.

Illumina, as part of the DCPI, is a platform to hold city administrations accountable for not only flawed design, but also the sustained failure to allot taxpayer funds to fix our cities. Data does not lie.

Bigyan nating liwanag ang ating mga siyudad tungo sa makataong disenyo.





A computer vision-powered internet-of-things device to detect industrial waste and protect crucial water bodies.

[Dashboard GitHub](#) | [Dashboard Prototype \(initial\)](#) | [Mobile App GitHub](#) | [Mobile App Prototype \(initial\)](#)

Authors: [Norman Marfa III](#), [Joemil Mendoza](#), [Rafael Zerna](#), [Simonee Mariquit](#), [Lawrence Macatangay](#)
from UPLB

Problem

Laguna Lake, an important source of freshwater surrounded by many cities and towns, faces serious pollution caused by untreated waste flowing from its connected rivers. This "bad water" (as local fishermen call it) results in fish dying and harms the lake's natural balance, putting at risk the incomes of people who depend on the lake for their living. Many pollution events are not reported because there are not enough good tools to monitor the water, and because people who rely on the lake often face urgent economic needs.

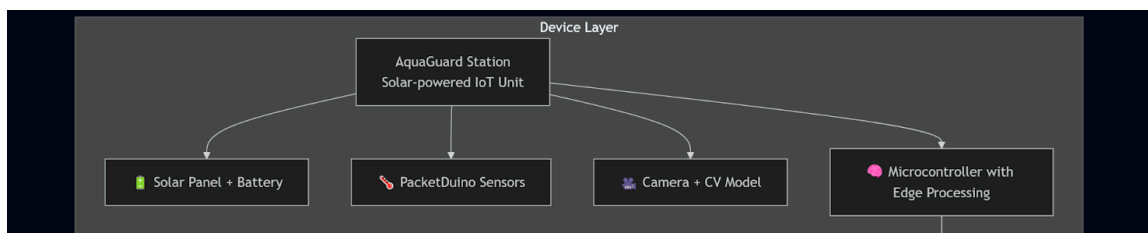
Background Solution (Documentation)

AquaGuard is a low-cost, solar-powered IoT system that monitors water quality in key river inlets of Laguna Lake. It continuously measures turbidity, dissolved oxygen, and color anomalies, delivering real-time data on water conditions. When pollution exceeds safe limits, the system sends instant alerts to authorities and nearby communities for rapid response.

It also includes a mobile app that lets residents report visible pollution through photos and geotags—enabling early detection of illegal waste dumping and encouraging public participation. By combining automated monitoring, community engagement, and data-driven insights, AquaGuard supports evidence-based environmental policy and fosters stewardship of Laguna Lake.

This initiative directly contributes to SDG 6 (Clean Water) and SDG 11 (Sustainable Communities).

Conceptual Framework of IoT Device

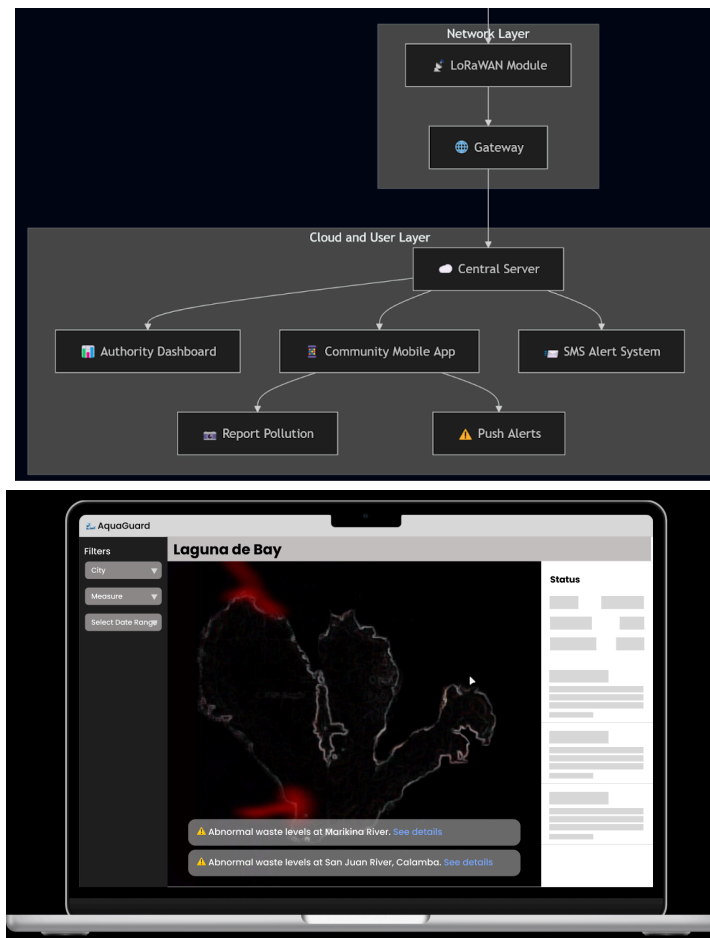


The AquaGuard device would be a compact, robust unit deployed at the mouths of connected Laguna Lake rivers.

- **Solar Panel:** Powers the device and charges an internal battery for continuous operation.

- **Sensors:** Integrated probes for turbidity, dissolved oxygen, and optical sensors for color anomaly detection.
- **Microcontroller/Processor:** Collects data from sensors, processes it, and prepares it for transmission.
- **Communication Module:** Transmits data wirelessly to a central server.
- **Enclosure:** Weatherproof and durable, designed to withstand environmental conditions.

Dashboard



The application will also implement a central dashboard for authorities and a user-friendly mobile application for the residents:

- **Authority Dashboard:**
 - **Interactive Map:** Displays the location of all AquaGuard devices and pollution hotspots.
 - **Real-time Data Streams:** Graphs and charts showing current and historical water quality parameters.
 - **Alert Log:** A clear list of all triggered alerts, severity, and timestamps.
 - **Incident Management:** Tools to track reported incidents and their resolution, allowing to create statistics and identify trends over time
- **Community Mobile App:**

- **"Report Pollution" Button:** Allows users to take a photo, add a description, and geotag the location.
- **Lake Health Status:** A simplified view of overall lake health or specific areas.
- **Educational Resources:** Information on pollution, its impacts, and how to help.
- **Notifications:** Alerts for major pollution events in their vicinity.

Feasibility

AquaGuard is highly feasible through partnerships with existing institutions. LGUs will provide on-ground support and integrate the system into waste management plans. The LLDA will aid in data sharing, policy alignment, and official endorsement. Support from the DENR ensures alignment with national policies and opens doors to technical guidance and funding. Collaborations with institutions like UP Marine Science Institute will offer expertise for sensor calibration and analysis, while community groups will drive awareness and mobile reporting success.

By combining technology, community engagement, and institutional support, AquaGuard offers a scalable, data-driven solution to protect Laguna Lake, safeguard livelihoods, and promote sustainable development.

References

JFE Advantech Devices — [DOI:10.1016/j.jglr.2021.03.017](https://doi.org/10.1016/j.jglr.2021.03.017)

IoT Device Studies:

- [10.1109/ICIIIECS.2015.7193080](https://doi.org/10.1109/ICIIIECS.2015.7193080)
- AWMC IoT Sensor Review
- [Systematic Review – ScienceDirect](#)

Laguna Lake Research:

- *Microplastics in Laguna Lake* – Environmental Science & Pollution Research
- *Heavy Metals in Sediments* – [MJST](#)
- *Water Quality & Trophic State* – [DOI:10.1080/1463498029003181](https://doi.org/10.1080/1463498029003181)



Resources

UI/UX

https://youtu.be/To_ADCVSg5g?si=VBcZm2zF7AstcDt8

▶ MASTER Figma Components in 10 Minutes (Everything You Need To Know)

▶ Learn Figma In Under 10 Minutes! (LEARN FIGMA FAST)

Developer Tools

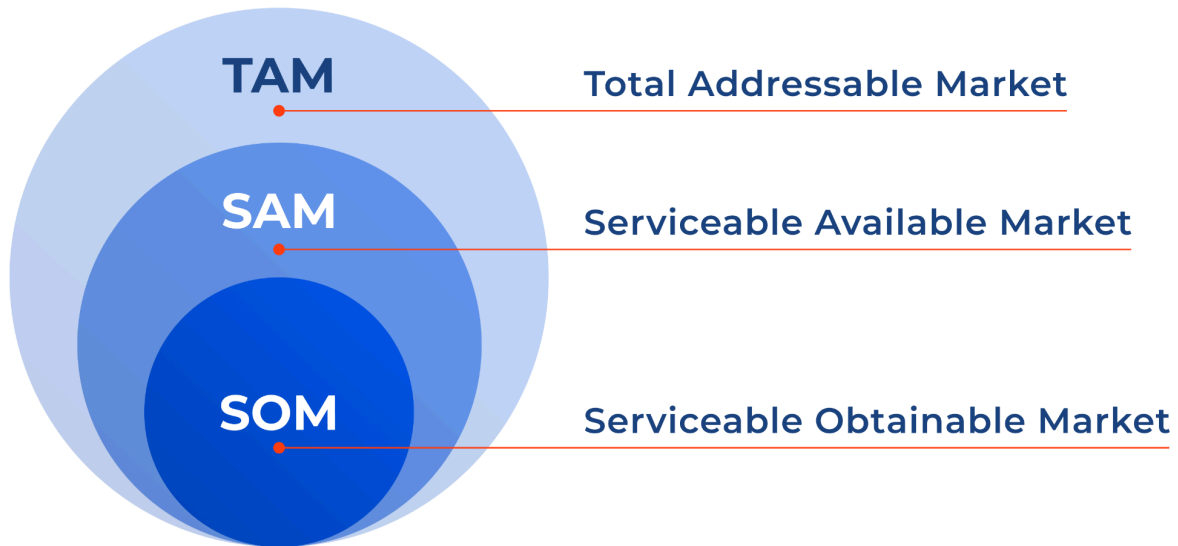
Software

Web

Mobile

Data

Business



The SCAMPER model

Seven perspectives to provoke creative solutions to challenging problems.

S	C	A	M	P	E	R
Substitute	Combine	Adapt	Modify	Put to another use	Eliminate	Reverse
Replace a part, material, or process with something else.	Join elements, ideas, or functions together in new ways – or find a new element you can merge with.	Modify something to better suit a new purpose, person or context.	Enlarge, reduce, change the shape, or alter attributes. Can a small change have a big effect?	Rather than changing the thing itself, consider changing the context it exists in.	Remove elements, simplify, or pare down to essentials. Is less more?	Flip the script, re-order your priorities, invert cause and effect, and turn it all upside-down.

Hackathon Info

▶ How to Win EVERY Hackathon (from a Top 50 Hacker)

▶ Hackathons, explained for beginners

<https://www.youtube.com/watch?v=lyLZukPGWn4>

Pitching

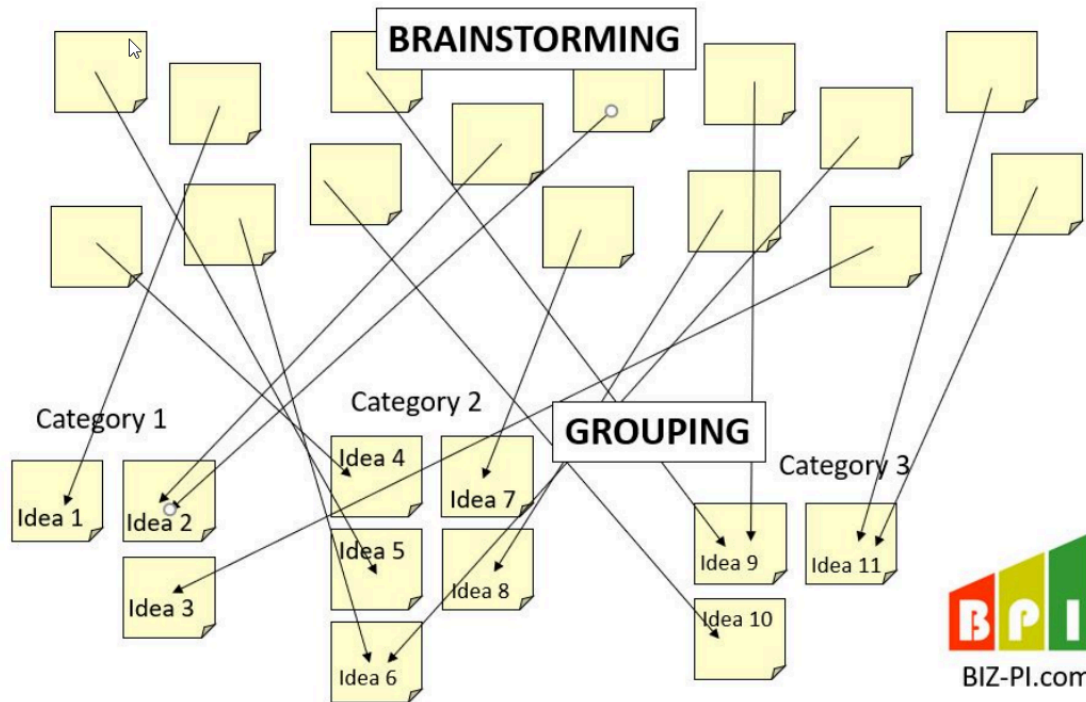
 The Winning Pitch of EOS Hackathon London Awarded \$100,000

Hardware

Ha

Ideation/PM

AFFINITY DIAGRAM (KJ METHOD)



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